# DIVISIBILITY OF CONCATENATED ARITHMETIC PROGRESSIONS AND CHAMPERNOWNE PREFIXES

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Abstract. We study divisibility questions for integers obtained by concatenation in positional notation. For fixed digit length, concatenating an arithmetic progression is an affine function of its initial term, which yields an exact congruence criterion. We use this to prove a global existence theorem in every even base; in decimal this resolves the existence conjectures recorded in OEIS A075001 and A075000.

We then consider the Champernowne prefixes

$$P(k) = 1 \parallel 2 \parallel \cdots \parallel k.$$

Our main theorem proves that for every positive integer $n$ there exists $k$ such that $n \mid P(k)$, thereby resolving the well-definedness conjecture in OEIS A075002. The proof has two local ingredients. For primes $p \notin \{2, 3, 5\}$, fixed-digit blocks admit unit translations and explicit state periods. The base primes 2 and 5 are handled by the terminal block, while 3 requires a separate block-shift lifting argument. The local roots are synchronized at one common prefix index by processing odd prime factors in increasing order; every completed state period has no prime divisor larger than the prime just processed.

## 1. Introduction

Let

$$P(k) = 1 \parallel 2 \parallel \cdots \parallel k$$

denote the decimal integer obtained by concatenating $1, 2, \ldots, k$, and for $n \geq 1$ let

$$C_n(k) = k \parallel (k+1) \parallel \cdots \parallel (k+n-1).$$

OEIS A075001 asks for the smallest $k$ such that $n \mid C_n(k)$, with value 0 if no such $k$ exists, while OEIS A075000 records the corresponding smallest quotient [4, 3]. OEIS A075002 asks for the smallest $k$ such that $n \mid P(k)$, where $P$ is also the sequence in OEIS A007908 [5, 2]. On September 11, 2026, the three entries still recorded these existence statements as conjectures. We are not aware of a previous proof of the full A075002 assertion; this paper makes no claim of priority.

Our first result is a general even-base construction for concatenated arithmetic progressions. The second and main result is the complete existence theorem for Champernowne prefixes. The proof strategy is modular. We

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first choose a single sufficiently long digit block. The endpoint is made divisible by the factor of the modulus whose prime divisors divide the base, and the exceptional 3-power is installed by a three-way block shift. The remaining odd prime powers are then installed in increasing order, using state periods to preserve all earlier congruences. The fact that a completed state period has no prime divisor larger than the prime already processed ensures that the next translation remains a unit; a finite capacity bound keeps every chosen endpoint in the same digit block.

Concatenation of arithmetic progressions in arbitrary bases has also been studied by Luca and Teguia Tabuguia, who derive explicit formulae and prove non-holonomicity [1]. Our emphasis is different: divisibility by an externally prescribed modulus and, in particular, the three OEIS existence problems above.

## 2. Fixed-digit arithmetic progressions

Let $\beta \geq 2$, let $n \geq 1$, let $s \in \mathbb{Z}$, and let $d \geq 1$. Put $B = \beta^d$. Suppose that

$$k,\ k+s,\ \ldots,\ k+(n-1)s$$

are positive $d$-digit base-$\beta$ integers. Their concatenation is

$$C_{\beta,n,s,d}(k) = \sum_{j=0}^{n-1} (k+js) B^{\,n-1-j}. \tag{1}$$

Define

$$R_n(B) = \sum_{i=0}^{n-1} B^i, \qquad T_n(B) = \sum_{j=0}^{n-1} j B^{\,n-1-j}.$$

Then

$$C_{\beta,n,s,d}(k) = kR_n(B) + sT_n(B). \tag{2}$$

**Proposition 2.1** (Fixed-digit affine congruence criterion)**.** *Under the fixed-digit hypotheses above, let $g = \gcd(R_n(B), n)$. The congruence in the affine expression (2), viewed as a congruence in $k \in \mathbb{Z}$,*

$$n \mid C_{\beta,n,s,d}(k)$$

*is solvable in $k$ if and only if*

$$g \mid sT_n(B).$$

*When solvable, the solutions form one residue class modulo $n/g$. This congruence criterion concerns solutions in $\mathbb{Z}$; admissibility of a representative in a particular finite fixed-digit interval is a separate condition.*

*Proof.* By (2), the desired divisibility is the linear congruence

$$kR_n(B) \equiv -sT_n(B) \pmod{n}.$$

The standard solvability criterion for a linear congruence gives the claim. After division by $g$, the coefficient $R_n(B)/g$ is invertible modulo $n/g$. □

**Theorem 2.2** (Even-base AP divisibility)**.** *Let $\beta \geq 2$ be even. For every $n \geq 1$ and every $s \in \mathbb{Z}$, there exist $d \geq 1$ and $k$ such that all*

$$k,\ k+s, \ldots, k+(n-1)s$$

*are positive $d$-digit base-$\beta$ integers and*

$$n \mid C_{\beta,n,s,d}(k).$$

*Proof.* Write

$$n = qm, \tag{3}$$

where $q$ is the largest divisor of $n$ all of whose prime divisors divide $\beta$. Thus $(q, m) = 1$ and $(m, \beta) = 1$.

Choose $d$ arbitrarily large subject to

$$q \mid \beta^d, \qquad \beta^d \equiv 1 \pmod{m}.$$

Such $d$ exist: the first condition holds for all sufficiently large $d$, and the second holds whenever $d$ is a multiple of $\mathrm{ord}_m(\beta)$, with no order condition when $m = 1$.

Put $B = \beta^d$, and let

$$u_- = \min\{0, (n-1)s\}, \qquad u_+ = \max\{0, (n-1)s\}.$$

All blocks have $d$ digits whenever

$$\beta^{d-1} - u_- \leq k \leq \beta^d - 1 - u_+. \tag{4}$$

The length of this interval tends to infinity with $d$, so we may choose $k$ in it such that

$$k + (n-1)s \equiv 0 \pmod{q}.$$

Modulo $q$, every term of the sum in (1) except the final block is multiplied by a positive power of $B$. Hence

$$C_{\beta,n,s,d}(k) \equiv k + (n-1)s \equiv 0 \pmod{q}. \tag{5}$$

Modulo $m$, since $B \equiv 1$,

$$C_{\beta,n,s,d}(k) \equiv nk + s\frac{n(n-1)}{2} \pmod{m}. \tag{6}$$

The first term is divisible by $m$. If $n$ is odd, then

$$s\frac{n(n-1)}{2} = mqs\frac{n-1}{2}.$$

If $n$ is even, then $2 \mid \beta$, so the full 2-part of $n$ lies in $q$; hence $q/2 \in \mathbb{Z}$, and

$$s\frac{n(n-1)}{2} = m\frac{q}{2}\, s(n-1).$$

Thus the concatenation is 0 modulo both coprime factors $q$ and $m$, hence modulo $n$. □

**Corollary 2.3** (A075001)**.** *For every $n \geq 1$ there exists a positive integer $k$ such that*

$$n \mid k \parallel (k+1) \parallel \cdots \parallel (k+n-1).$$

*Hence OEIS A075001 has no zero terms.*

*Proof.* Take $\beta = 10$ and $s = 1$ in Theorem 2.2. □

**Corollary 2.4** (A075000)**.** *For every $n \geq 1$, OEIS A075000 is nonzero. More precisely,*

$$A075000(n) = \frac{C_n(A075001(n))}{n}.$$

*Proof.* Existence follows from Corollary 2.3. It remains only to justify minimality. For fixed $n$, if $k_2 > k_1$, every term in the tuple for $k_2$ is larger than the corresponding term for $k_1$, so every corresponding digit length is at least as large. If the total concatenated lengths differ, $C_n(k_2)$ is longer and hence numerically larger; if they agree, the coordinatewise length inequalities force equality of every corresponding block length, and the first position where the concatenated strings differ lies in an equal-length block whose decimal representation is larger for $k_2$. Thus $C_n(k_2) > C_n(k_1)$, including across powers of ten. The least divisible concatenation therefore occurs at the least witnessing index $A075001(n)$, and division by the fixed positive integer $n$ preserves the ordering. □

## 3. Fixed-block identities for Champernowne prefixes

Fix a digit length $d$ and put $B = 10^d$. Let $k \geq 1$ and $L \geq 0$, and suppose that every integer in the appended interval $k+1, \ldots, k+L$ remains in this same $d$-digit block (with the condition vacuous when $L = 0$). Then

$$P(k+L) = B^L P(k) + \sum_{j=1}^{L} (k+j) B^{L-j}. \tag{7}$$

Set

$$S_L = \sum_{r=0}^{L-1} B^r, \qquad T_L = \sum_{j=1}^{L} j B^{L-j}.$$

Then

$$P(k+L) - P(k) = (B^L - 1)P(k) + kS_L + T_L. \tag{8}$$

A direct telescoping calculation gives

$$(B-1)T_L = BS_L - L. \tag{9}$$

## 4. Local lifting and state periods

**Lemma 4.1** (Unit translation)**.** *Fix a $d$-digit block and put $B = 10^d$. Let $p$ be an odd prime with $p \nmid B(B-1)$, and let*

$$h = \operatorname{ord}_p(B), \qquad U = T_h,$$

*let $a \geq 0$, and let $c \geq 1$ with $p \nmid c$. Put $L = chp^a$. If $k \geq 1$ and every integer $k+1, \ldots, k+L$ remains in the same fixed $d$-digit block, then*

$$P(k+L) - P(k) \equiv cp^a U \pmod{p^{a+1}}, \tag{10}$$

*Moreover $p \nmid U$.*

*Proof.* Modulo $p$, $B^h = 1$ and $S_h = 0$. From (9),

$$(B-1)U \equiv -h \pmod{p}.$$

Since $h \mid p-1$, both $h$ and $B-1$ are units modulo $p$, hence so is $U$.

By the lifting-the-exponent lemma,

$$B^{chp^a} \equiv 1 \pmod{p^{a+1}}.$$

Since $B-1$ is a unit, this also gives

$$S_L \equiv 0 \pmod{p^{a+1}}.$$

Equation (9) therefore yields

$$T_L \equiv -\frac{chp^a}{B-1} \equiv cp^a U \pmod{p^{a+1}}.$$

Substitution into (8) proves (10). □

**Lemma 4.2** (Fixed-block state period)**.** *Fix a $d$-digit block and put $B = 10^d$. Let $p$ be an odd prime with $p \nmid B(B-1)$, and let $e \geq 1$. With*

$$H = p^e \operatorname{ord}_{p^e}(B) \tag{11}$$

*and $k \geq 1$, if every integer $k+1, \ldots, k+H$ remains in the same fixed $d$-digit block, then $H$ is a period of the endpoint state modulo $p^e$:*

$$P(k+H) \equiv P(k) \pmod{p^e}, \qquad k+H \equiv k \pmod{p^e}. \tag{12}$$

*More generally, for every $m \geq 1$, the same two congruences with $H$ replaced by $mH$ hold provided every integer $k+1, \ldots, k+mH$ remains in that same fixed $d$-digit block. Every prime divisor of $H$ is at most $p$.*

*Proof.* Let $r = \operatorname{ord}_{p^e}(B)$. Since $r \mid H$, $B^H \equiv 1 \pmod{p^e}$. Because $B-1$ is a unit, $S_H \equiv 0 \pmod{p^e}$. Also $p^e \mid H$, so (9) gives $T_H \equiv 0 \pmod{p^e}$. Equation (8) proves the first congruence in (12), and $p^e \mid H$ proves the second. If $m \geq 1$ and the longer appended interval remains in the block, the same calculation applies with $mH$ in place of $H$: $r \mid mH$, $p^e \mid mH$, and hence $P(k+mH) \equiv P(k) \pmod{p^e}$ and $k+mH \equiv k \pmod{p^e}$.

Finally,

$$\operatorname{ord}_{p^e}(B) \mid \varphi(p^e) = p^{e-1}(p-1),$$

so every prime divisor of $H$ is at most $p$. □

**Lemma 4.3** (Exceptional 3-adic block shift)**.** *Fix $d \geq 1$, put $B = 10^d$, and let $L \geq 1$ satisfy $v_3(L) = a \geq 1$. If $k \geq 1$ and every integer $k+1, \ldots, k+L$ remains in the same fixed $d$-digit block, then*

$$P(k+L) - P(k) \equiv Lk + \frac{L(L+1)}{2} \pmod{3^{a+1}} \tag{13}$$

*Proof.* Write $B = 1 + c$. Since $B = 10^d$, we have $9 \mid c$. Expansion of the geometric sum gives

$$S_L - L = \sum_{s \geq 1} c^s \binom{L}{s+1}.$$

For $m = s + 1$,

$$\binom{L}{m} = \frac{L}{m}\binom{L-1}{m-1},$$

so

$$v_3\binom{L}{m} \geq a - v_3(m)$$

and $v_3(c) \geq 2$. For $s \geq 1$, $v_3(s+1) \leq s$, so every nonzero correction term on the right has valuation at least $2s + a - v_3(s+1) \geq a + 1$. Hence

$$S_L \equiv L \pmod{3^{a+1}}.$$

Similarly,

$$T_L - \frac{L(L+1)}{2} = \sum_{s \geq 1} c^s \binom{L+1}{s+2}.$$

From

$$\binom{L+1}{m} = \frac{L(L+1)}{m(m-1)}\binom{L-1}{m-2}$$

we obtain

$$v_3\left(c^s \binom{L+1}{s+2}\right) \geq 2s + a - v_3((s+1)(s+2)).$$

For $s \geq 1$,

$$v_3((s+1)(s+2)) \leq 2s - 1;$$

for $s = 1$ this is immediate, and for $s \geq 2$ it follows from $3^{2s} > (s+1)(s+2)$. Thus every correction term is divisible by $3^{a+1}$.

Finally LTE gives $B^L \equiv 1 \pmod{3^{a+2}}$, so the first term in (8) vanishes modulo $3^{a+1}$. Substituting the two congruences for $S_L$ and $T_L$ into (8) proves (13). □

**Corollary 4.4** (3-adic lifting)**.** *Fix a $d$-digit block with $B = 10^d$. Let $R = 2^\alpha 5^\beta$, let $k \geq 1$, suppose $R \mid k$ and $3^a \mid P(k)$ with $a \geq 1$, and put*

$$L = \operatorname{lcm}(2 \cdot 3^a, R) = 2c3^a, \qquad 3 \nmid c. \tag{14}$$

*Assume that every integer $k+1, \ldots, k+2L$ remains in the same fixed $d$-digit block. Then exactly one of*

$$k, \qquad k + L, \qquad k + 2L$$

*is a root modulo $3^{a+1}$, while all three endpoint indices remain divisible by $R$. Moreover*

$$2 \cdot 3^e \tag{15}$$

*is a state period modulo $3^e$ for every $e \geq 1$, provided the relevant appended interval remains in the same fixed digit block. More generally, every positive multiple of $2 \cdot 3^e$ is such a state period modulo $3^e$ when its entire appended interval remains in that block.*

*Proof.* For $t = 0$, the desired quotient is 0. For $t = 1, 2$, we have $v_3(tL) = a$, and the preceding lemma applies to the full interval $k+1, \ldots, k+2L$. It gives

$$P(k+tL) - P(k) \equiv tLk + \frac{tL(tL+1)}{2} \pmod{3^{a+1}}.$$

The right-hand side is divisible by $3^a$, so division by $3^a$, followed by reduction modulo 3, gives for all $t = 0, 1, 2$

$$\frac{P(k+tL) - P(k)}{3^a} \equiv tc(2k+1) \pmod 3. \tag{16}$$

Since $3 \mid P(k)$ and $10 \equiv 1 \pmod 3$,

$$0 \equiv P(k) \equiv 1 + 2 + \cdots + k = \frac{k(k+1)}{2} \pmod 3,$$

so $k \equiv 0$ or $2 \pmod 3$. Hence $c(2k+1)$ is a unit modulo 3. Therefore

$$\frac{P(k+tL)}{3^a} \equiv \frac{P(k)}{3^a} + tc(2k+1) \pmod 3,$$

and $t = 0, 1, 2$ gives all three residue classes modulo 3, so exactly one choice is a root modulo $3^{a+1}$. Because $R \mid L$, all three endpoint indices remain divisible by $R$.

For the period assertion, let $m \geq 1$ and take $L = m(2 \cdot 3^e)$ in Lemma 4.3. Its valuation is at least $e$, so the resulting congruence implies $P(k+L) \equiv P(k) \pmod{3^e}$, while $k + L \equiv k \pmod{3^e}$. This application is valid only when the entire appended interval through $k + L$ stays inside the fixed block. □

## 5. A common prefix for every modulus

**Theorem 5.1** (A075002)**.** *For every positive integer $n$, there exists $k \geq 1$ such that*

$$n \mid P(k).$$

*Consequently OEIS A075002 is well-defined for every $n$.*

*Proof.* For $n = 1$, take $k = 1$. Hence assume $n > 1$. Factor

$$n = R\,3^e \prod_{i=1}^{r} p_i^{e_i}, \qquad R = 2^\alpha 5^\beta, \qquad 7 \leq p_1 < \cdots < p_r. \tag{17}$$

Let

$$D = \operatorname{lcm}_{1 \leq i \leq r} \operatorname{ord}_{p_i^{e_i}}(10),$$

with $D = 1$ if $r = 0$. We shall work in one sufficiently long $d$-digit block, where

$$d \equiv 1 \pmod{D}, \qquad d \geq \max(\alpha, \beta). \tag{18}$$

Such $d$ are arbitrarily large. Put $B = 10^d$. Then for every $i$,

$$B \equiv 10 \pmod{p_i^{e_i}}.$$

For the state periods and the step lengths below, define

$$H_0 = \begin{cases} 2 \cdot 3^e, & e > 0, \\ 1, & e = 0. \end{cases} \tag{19}$$

For $1 \leq i \leq r$, define

$$H_i = \operatorname{lcm}\Big(H_{i-1},\ p_i^{e_i} \operatorname{ord}_{p_i^{e_i}}(10)\Big). \tag{20}$$

For $1 \leq i \leq r$, put

$$\begin{aligned} J_{i,0} &= \operatorname{lcm}(\operatorname{ord}_{p_i}(10), H_{i-1}, R), \\ J_{i,a} &= \operatorname{lcm}(\operatorname{ord}_{p_i}(10) p_i^a, H_{i-1}, R) \quad (1 \leq a < e_i). \end{aligned} \tag{21}$$

Empty sums below are understood to be zero. An explicit bound for the total forward displacement of all candidate endpoints is

$$W(n) = 2 \sum_{a=1}^{e-1} \operatorname{lcm}(2 \cdot 3^a, R) + \sum_{i=1}^{r} (p_i - 1) \sum_{a=0}^{e_i - 1} J_{i,a}. \tag{22}$$

Indeed, a 3-adic stage uses one of $tL_a$, $0 \leq t \leq 2$, and hence contributes at most $2L_a$. At a $p_i$-adic stage, one of $tJ_{i,a}$, $0 \leq t < p_i$, is used, and hence contributes at most $(p_i - 1)J_{i,a}$. Thus $W(n)$ bounds not only the selected trajectory but also every candidate endpoint examined at every stage. All quantities in (19)–(22) depend only on $n$. Indeed, $d \equiv 1 \pmod{D}$ implies

$$10^d \equiv 10 \pmod{p_i^{e_i}},$$

so the orders of $B = 10^d$ required by the state-period lemma are exactly the corresponding orders of 10.

Let

$$M_0 = R \cdot 3^{\min(e,1)}. \tag{23}$$

Choose $d$ in the congruence class in (18) so large that

$$9 \cdot 10^{d-1} > M_0 + W(n). \tag{24}$$

Let $k_0$ be the least multiple of $M_0$ that is at least $10^{d-1}$. Then $0 \leq k_0 - 10^{d-1} < M_0$. Every candidate endpoint in the construction is at most $k_0 + W(n)$, and therefore

$$k_0 + W(n) < 10^{d-1} + M_0 + W(n) < 10^{d-1} + 9 \cdot 10^{d-1} = 10^d.$$

All candidate endpoints are also at least $k_0 \geq 10^{d-1}$. Since every appended interval lies between its current endpoint and one of these candidate endpoints, this single capacity estimate supplies the fixed-block hypothesis for every translation, state-period, and 3-adic block-shift invocation below.

Since $R \mid k_0$ and $B \equiv 0 \pmod{R}$,

$$P(k_0) = BP(k_0 - 1) + k_0 \equiv 0 \pmod{R}. \tag{25}$$

If $e > 0$, then $3 \mid k_0$, and because $10 \equiv 1 \pmod{3}$,

$$P(k_0) \equiv \frac{k_0(k_0+1)}{2} \equiv 0 \pmod{3}.$$

If $e > 1$, repeatedly apply Corollary 4.4. At stage $a$, use

$$L_a = \operatorname{lcm}(2 \cdot 3^a, R).$$

The capacity estimate above puts the full interval through $k + 2L_a$ in the fixed block, so the corollary applies to all three candidates. One of the three choices raises a root modulo $3^a$ to a root modulo $3^{a+1}$. Since $R \mid L_a$ and $B \equiv 0 \pmod{R}$, the transition identity gives $P(k + tL_a) \equiv k + tL_a \equiv 0 \pmod{R}$ for every $t > 0$, while $t = 0$ is the current root. Thus the $R$-condition is preserved and we obtain an endpoint $k$ with

$$R3^e \mid P(k).$$

If $e > 0$, Corollary 4.4 supplies the conditional fixed-block state period $H_0 = 2 \cdot 3^e$ modulo $3^e$; if $e = 0$, there is no 3-adic condition to preserve.

Now process $p_1, \ldots, p_r$ in increasing order. Suppose all preceding prime powers have been installed at the current endpoint $k$, and $H_{i-1}$ is a common conditional fixed-block state period for them. Let $p = p_i$ and

$$h = \operatorname{ord}_p(10).$$

Since $B \equiv 10 \pmod{p}$, we also have $h = \operatorname{ord}_p(B)$, and $p \nmid B(B-1)$.

Every prime factor of the period contributed by an earlier prime $q < p$ is at most $q$, while $R$ has no prime divisors other than 2 and 5. Therefore $p \nmid H_{i-1}R$, and also $p \nmid h$ since $h \mid p - 1$. Hence

$$J_{i,0} = c_0 h, \qquad p \nmid c_0.$$

Let $U = T_h$. For $t = 0$ the following congruence is trivial; for $1 \leq t < p$, apply Lemma 4.1 with $c = tc_0$ and $L = tJ_{i,0}$. The complete interval through the largest candidate is in the fixed block by the capacity estimate, so

$$P(k + tJ_{i,0}) - P(k) \equiv tc_0 U \pmod{p}, \qquad 0 \leq t < p. \tag{26}$$

Since $p \nmid U$, the map

$$P(k) + tc_0 U, \qquad t \in \mathbb{F}_p,$$

runs through all residues modulo $p$: it is an affine permutation of $\mathbb{F}_p$. Because $J_{i,0}$ is a multiple of $H_{i-1}$, the fixed-block state-period lemma preserves every previously installed prime-power condition for every candidate $tJ_{i,0}$. Also $R \mid J_{i,0}$, and $B \equiv 0 \pmod{R}$; whenever $R \mid P(k)$, the transition identity

gives $P(k + tJ_{i,0}) \equiv k + tJ_{i,0} \equiv 0 \pmod{R}$ for $t > 0$, while $t = 0$ is the current root. Choose the unique $t$ giving a root modulo $p$.

Suppose inductively that a root modulo $p^a$ has been obtained, where $1 \leq a < e_i$. Again $p \nmid H_{i-1}Rh$, so $J_{i,a} = c_a h p^a$ with $p \nmid c_a$. For $1 \leq t < p$, Lemma 4.1 applied to $tJ_{i,a}$, with the fixed-block hypothesis supplied by the capacity estimate, gives

$$P(k + tJ_{i,a}) - P(k) \equiv tc_a p^a U \pmod{p^{a+1}}.$$

The congruence is also trivial for $t = 0$, so it holds for $0 \leq t < p$. Since $P(k) \equiv 0 \pmod{p^a}$, division by $p^a$ gives

$$\frac{P(k + tJ_{i,a})}{p^a} \equiv \frac{P(k)}{p^a} + tc_a U \pmod{p}, \qquad 0 \leq t < p.$$

The coefficient $c_a U$ is a unit modulo $p$, so the $p$ candidate endpoints traverse all possible next $p$-adic digits and exactly one is a root modulo $p^{a+1}$. Because $J_{i,a}$ is a multiple of $H_{i-1}$ and $R$, the same fixed-block state-period and $R$-preservation arguments keep all earlier conditions intact. Repeating installs $p^{e_i}$.

After completing this prime power, Lemma 4.2 gives the conditional fixed-block state period

$$p_i^{e_i} \operatorname{ord}_{p_i^{e_i}}(B) = p_i^{e_i} \operatorname{ord}_{p_i^{e_i}}(10).$$

Consequently $H_i$ in (20) is a common conditional state period for all installed prime powers. Every prime divisor of the new factor is at most $p_i$, so the same argument applies at the next, larger prime.

At termination, the same endpoint $k$ satisfies every prime-power condition in the factorization (17). Therefore $n \mid P(k)$. □

*Remark* 5.2. The increasing-prime order is essential to this proof. A single global affine orbit need not have unit translation simultaneously at all prime factors. The construction above avoids that obstruction by preserving completed states with periods whose prime divisors are all at most the prime just processed; a larger, unprocessed prime therefore cannot divide the next step's coefficient.

## 6. Verification and provenance

The proofs do not use computation as a premise. A separate verifier evaluates prefixes modulo the requested modulus using fixed-block transition matrices and cross-checks each matrix jump against a literal recurrence on small instances. The recorded checks include the published initial values of A075002, the A075001 construction for $n \leq 2000$, both lifting lemmas, and common-index constructions for $n \leq 500$, together with the five composite moduli below 5000 for which a single affine orbit cannot provide unit translations at all prime factors. These computations are regression and reproducibility checks, not premises of the proofs.

The source, bibliography, verifier, verification record, and mathematical audit record are preserved in the public repository:

`https://github.com/dongxuelian2/oeis-papers`

The current release is

`a075000-a075002-v1.0.4`

`https://github.com/dongxuelian2/oeis-papers/releases/tag/a075000-a075002-v1.0.4`

It contains this paper as a stable snapshot. From the paper directory, the verifier can be reproduced with

```
python verify_concatenation_divisibility.py
```

The manuscript grew out of a larger computer-assisted investigation, but that research history is not used as a premise here. The OEIS status statement above records only the checked date; it is not intended as a claim of literature-wide priority.

## References


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